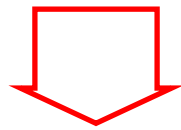


Recall:

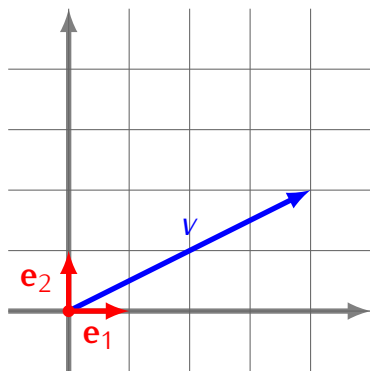
1) A basis $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ of a vector space V defines a coordinate system:

$$\mathbf{v} = c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n$$

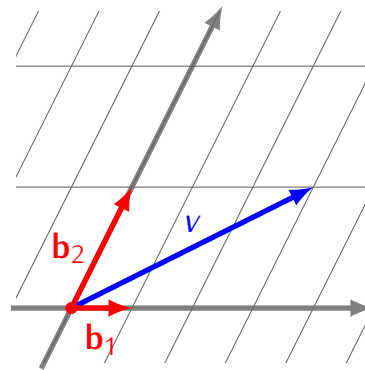


$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

2) Different bases define different coordinate systems.



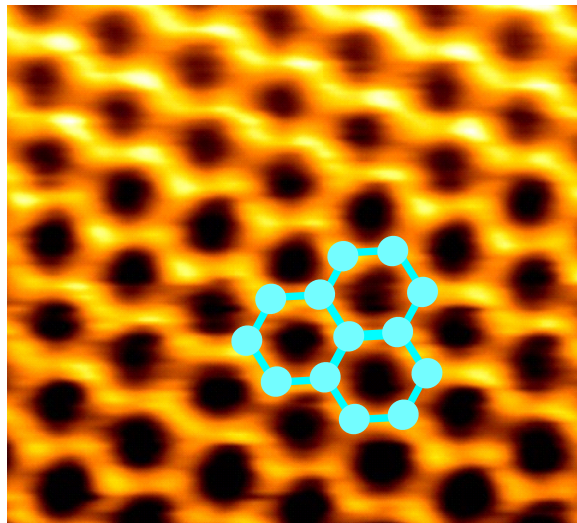
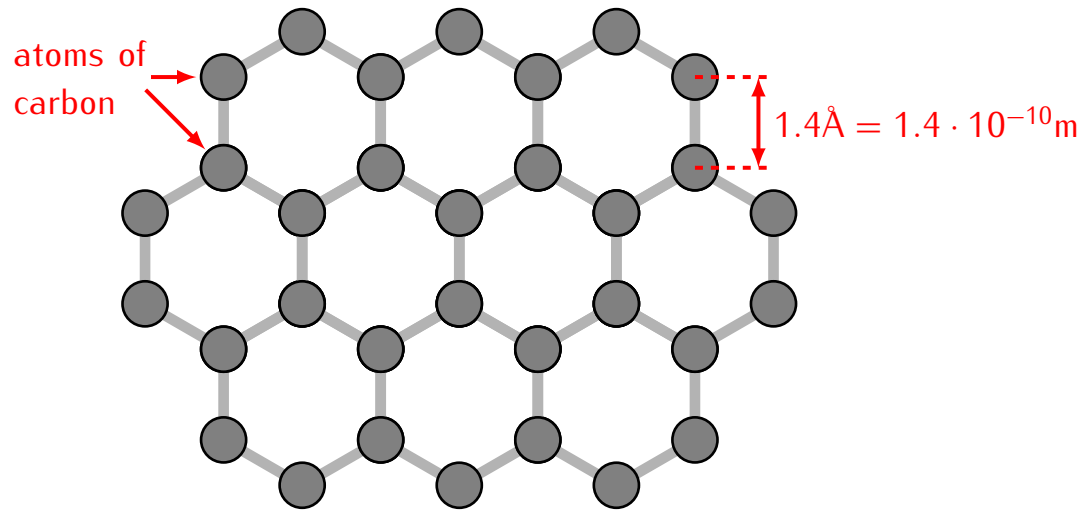
$$\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2\}$$



$$\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$$

Note. Choosing a convenient basis can simplify computations.

Example. Graphene lattice.



*Image of graphene taken with an atomic force microscope.
© University of Augsburg, Experimental Physics IV.*

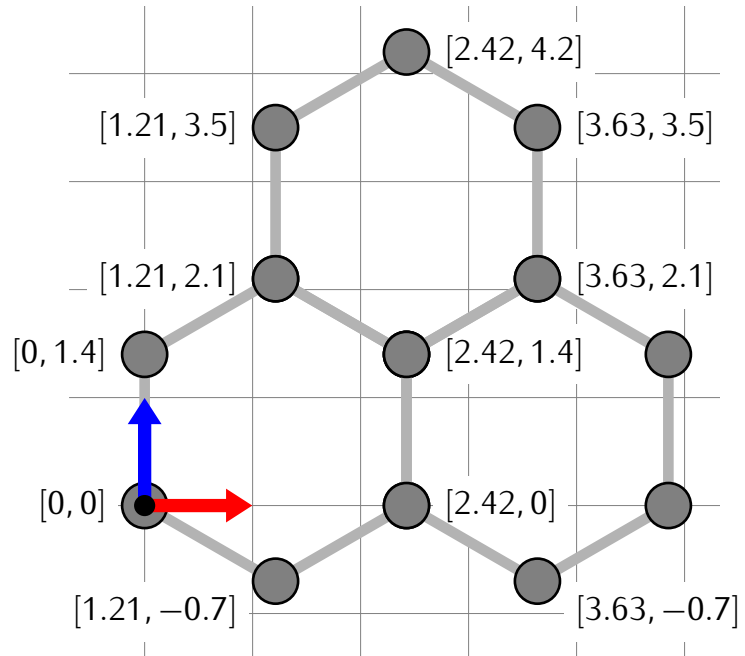
Coordinates of atoms in the graphene lattice

In the standard basis

$\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2\}$:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

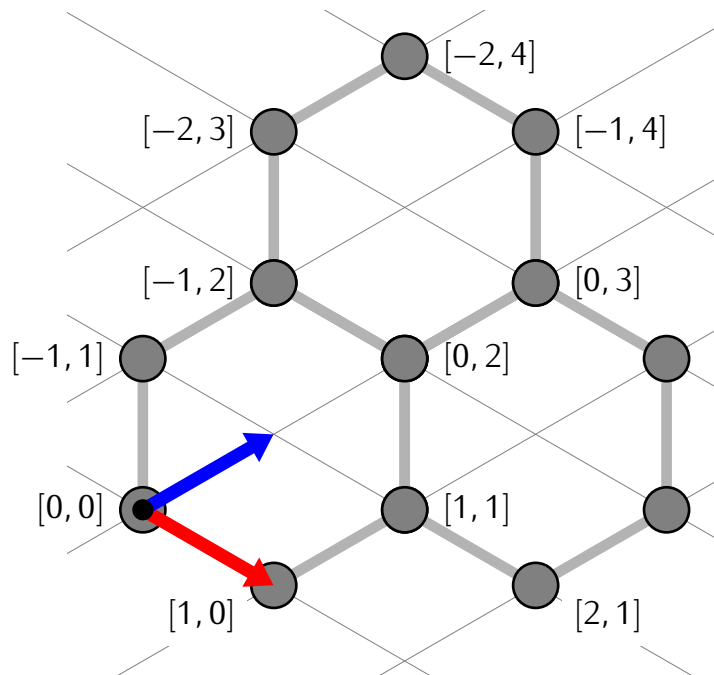
$$\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



In a more convenient
basis $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$:

$$\mathbf{b}_1 = \begin{bmatrix} 1.21 \\ -0.7 \end{bmatrix}$$

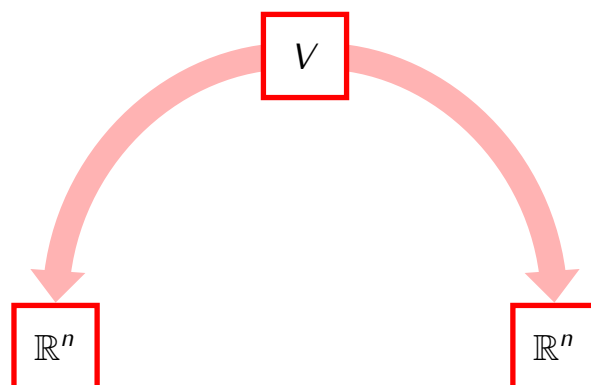
$$\mathbf{b}_2 = \begin{bmatrix} 1.21 \\ 0.7 \end{bmatrix}$$



Problem Let

$$\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}, \quad \mathcal{D} = \{\mathbf{d}_1, \dots, \mathbf{d}_1\}$$

be two bases of a vector space V , and let $\mathbf{v} \in V$. Assume that we know $[\mathbf{v}]_{\mathcal{B}}$. What is $[\mathbf{v}]_{\mathcal{D}}$?



Definition

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $\mathcal{D} = \{\mathbf{d}_1, \dots, \mathbf{d}_1\}$ be two bases of a vector space V . The matrix

$$P_{\mathcal{D} \leftarrow \mathcal{B}} = [[\mathbf{b}_1]_{\mathcal{D}} \quad [\mathbf{b}_2]_{\mathcal{D}} \quad \dots \quad [\mathbf{b}_n]_{\mathcal{D}}]$$

is called the *change of coordinates matrix* from the basis $\mathcal{B}$ to the basis $\mathcal{D}$.

Proposition

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $\mathcal{D} = \{\mathbf{d}_1, \dots, \mathbf{d}_1\}$ be two bases of a vector space V . For any vector $\mathbf{v} \in V$ we have

$$[\mathbf{v}]_{\mathcal{D}} = P_{\mathcal{D} \leftarrow \mathcal{B}} \cdot [\mathbf{v}]_{\mathcal{B}}$$

Example. Let $\mathbb{P}_2$ be the vector space of polynomials of degree ≤ 2 . Consider two bases of $\mathbb{P}_2$:

$$\mathcal{B} = \{1, 1 + t, 1 + t + t^2\}$$
$$\mathcal{D} = \{1 + t, 1 - 5t, 2 + t^2\}$$

1) Compute the change of coordinates matrix $P_{\mathcal{D} \leftarrow \mathcal{B}}$.

2) Let $p \in \mathbb{P}_2$ be a polynomial such that

$$[p]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$$

Compute $[p]_{\mathcal{D}}$.

Proposition

If $\mathcal{B}, \mathcal{D}, \mathcal{E}$ are three bases of a vector space V then:

1) $P_{\mathcal{B} \leftarrow \mathcal{D}} = (P_{\mathcal{D} \leftarrow \mathcal{B}})^{-1}$

2) $P_{\mathcal{E} \leftarrow \mathcal{B}} = P_{\mathcal{E} \leftarrow \mathcal{D}} \cdot P_{\mathcal{D} \leftarrow \mathcal{B}}$